Empirical Diagnostics of Adaptive Optimization: Replicating the Period-3 Convergence Trap in Adam*

Suryansh U.
Indian Institute of Technology (IIT) Delhi

October 16, 2026

Abstract

While the Adam optimizer dominates modern stochastic optimization, its formal convergence proof within the Online Convex Optimization (OCO) framework relies on a mathematically unfulfilled assumption. In environments characterized by sparse, highly informative gradient bursts, Adam's exponential moving average of squared gradients decays prematurely. This decay violates the positive semi-definiteness condition of the differential metric matrix ($\Gamma_t \succeq 0$), forcing divergence. We present a verified, from-scratch NumPy reproduction of this theoretical failure mechanism, expanding the diagnostic analysis across variance memory, parameter trajectories, and effective step-size inflation. The full implementation and testing suite are available at https://github.com/BinaryGenius06/adam-amsgrad-research.

1 Introduction

Adam [1] synthesizes the coordinate-wise scaling of AdaGrad [4] with the exponential moving averages (EMA) of RMSProp [5]. By heavily discounting older gradients, Adam adapts rapidly to non-stationary objective landscapes. However, Reddi et al. [2] demonstrated that Adam diverges on elementary one-dimensional convex objectives when exposed to infrequent but massive gradient bursts. The second-moment estimate decreases rapidly during non-burst iterations, causing the effective step size to artificially explode. AMSGrad resolves this by enforcing a monotonic running maximum on the raw second-moment accumulator.

*Code, figures, and the full `pytest` suite are available at https://github.com/BinaryGenius06/adam-amsgrad-research.

2 Theoretical Framework

2.1 OCO setting and regret

At each round $t = 1, \dots, T$, the learner selects $x_t \in \mathcal{F}$, where $\mathcal{F} \subset \mathbb{R}^d$ is a compact convex set, and then observes a convex loss f_t with gradient $g_t = \nabla f_t(x_t)$ [3]. Performance is measured by the regret

$$R_T = \sum_{t=1}^{T} f_t(x_t) - \min_{x \in \mathcal{F}} \sum_{t=1}^{T} f_t(x). \tag{1}$$

An algorithm converges in the OCO sense if $R_T = o(T)$, i.e. the average regret $R_T/T \to 0$.

2.2 Adam updates

With step size α_t, decay rates $\beta_1, \beta_2 \in [0, 1)$, and $m_0 = v_0 = 0$, Adam maintains

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1)\, g_t, \tag{2}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)\, g_t^2, \tag{3}$$

$$x_{t+1} = \Pi_{\mathcal{F}}\Big(x_t - \alpha_t \frac{m_t}{\sqrt{v_t}}\Big), \tag{4}$$

where $\Pi_{\mathcal{F}}$ is Euclidean projection onto $\mathcal{F}$ and all operations are coordinate-wise (bias correction is omitted, following the analysis in [2]). Unrolling Eq. (3) gives $v_t = (1-\beta_2)\sum_{s=1}^{t} \beta_2^{t-s} g_s^2$, so a large gradient at step s is forgotten geometrically at rate β_2.

2.3 The differential metric matrix

Writing the update as a preconditioned step with $V_t = \mathrm{diag}(v_t)$, the standard regret proof for adaptive methods controls the change in the effective inverse learning rate,

$$\Gamma_t \triangleq \frac{\mathrm{diag}(\sqrt{v_t})}{\alpha_t} - \frac{\mathrm{diag}(\sqrt{v_{t-1}})}{\alpha_{t-1}}. \tag{5}$$

The OCO proof requires $\Gamma_t \succeq 0$ for all t. This holds for AdaGrad, where v_t is a running sum and therefore

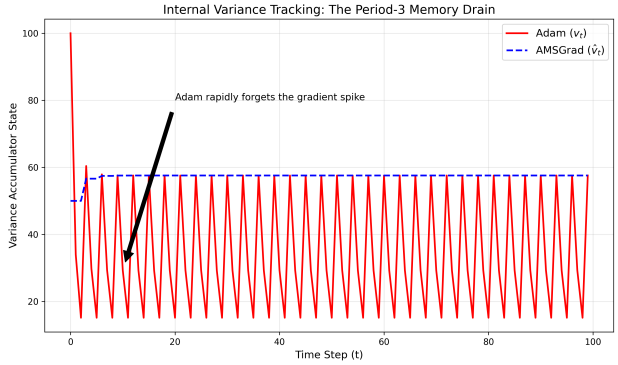


Figure 1: Adam (red) exhibits exponential memory decay during non-burst steps. AMSGrad (blue) enforces monotonicity.

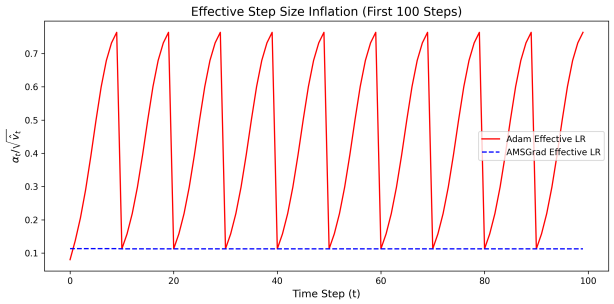


Figure 2: Effective step size inflation. Adam's step size (red) spikes rhythmically due to the decaying denominator.

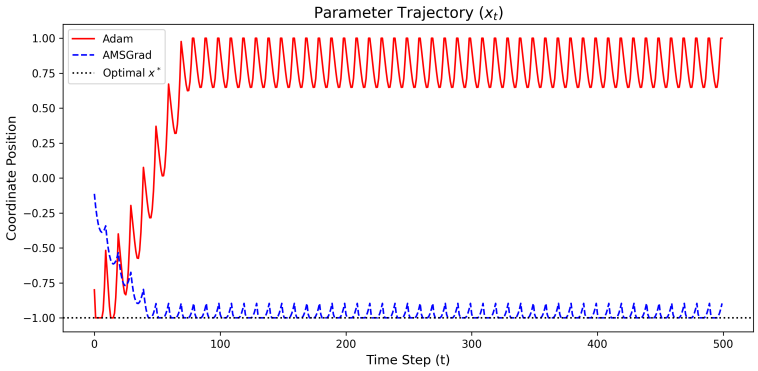


Figure 3: Optimization trajectory on $[-1, 1]$. Adam diverges to the sub-optimal boundary while AMSGrad converges.

non-decreasing. Because Adam computes v_t via an EMA, v_t shrinks whenever a large gradient is followed by smaller ones, generating negative entries in Γ_t and breaking the theoretical guarantee.

2.4 AMSGrad correction

AMSGrad [2] keeps Eqs. (2)–(3) but replaces the denominator with a running maximum:

$$\hat{v}_t = \max(\hat{v}_{t-1},\, v_t), \tag{6}$$

$$x_{t+1} = \Pi_{\mathcal{F}}\Big(x_t - \alpha_t \frac{m_t}{\sqrt{\hat{v}_t}}\Big). \tag{7}$$

Since $\hat{v}_t$ is non-decreasing by construction, $\Gamma_t \succeq 0$ is restored (with step size $\alpha_t = \alpha/\sqrt{t}$), and the standard analysis yields $R_T = O(\sqrt{T})$, hence $R_T/T \to 0$.

3 Diagnostic Reproduction

We implemented a 1D OCO counterexample constrained to $\mathcal{F} = [-1, 1]$, following [2]. The loss is

$$f_t(x) = \begin{cases} C\,x, & t \bmod 3 = 1, \\ -x, & \text{otherwise}, \end{cases} \tag{8}$$

so that every 3 steps the environment delivers a massive gradient $g_t = C = 10$, and for the remaining two steps $g_t = -1$. The cumulative loss over one period is $(C-2)x = 8x$, hence the globally optimal position is $x^* = -1$. Adam's EMA, however, over-weights the two small negative gradients once the burst has been forgotten, pulling x_t in the wrong direction. If $x_t \approx +1$ for all t, each period incurs $2(C-2) = 16$ excess loss, so $R_T \approx \frac{16}{3}T$ grows linearly.

3.1 Variance Memory Drain

Figure 1 demonstrates Adam's core mathematical failure point. During the non-burst steps, Adam's

variance accumulator (red) exponentially crashes toward zero, forgetting the prior burst. AMSGrad (blue) locks the maximum variance horizontally.

3.2 Effective Step-Size Inflation

Because the coordinate update is scaled by $1/\sqrt{v_t}$, the memory drain has a direct inversely proportional effect on the learning rate. As shown in Figure 2, every time Adam's variance crashes, its effective step size $\alpha_t/\sqrt{v_t}$ artificially spikes. This inflation occurs exactly when the gradient is $g_t = -1$, empowering the incorrect gradient to dominate the trajectory.

3.3 Parameter Divergence

The macro-level result of this inflation is catastrophic divergence. Figure 3 tracks the parameter x_t over 500 steps. Adam is quickly ratcheted to the sub-optimal boundary $+1$, completely failing the OCO objective. By maintaining a stable effective learning rate, AMSGrad correctly descends to the global optimum -1.

4 Validation and Conclusion

By establishing an automated `pytest` architecture, we confirmed strict mathematical parity between our custom NumPy engine and PyTorch's backend (tolerance 1×10^{-5}); Table 1 summarizes the checks. Exposing the internal states proves that Adam's divergence is an algorithmic phenomenon, not an implementation artifact. Enforcing the AMSGrad running maximum is strictly necessary to survive high-magnitude sparsity.

Table 1: Parity checks between the NumPy implementation and PyTorch (tol $= 1 \times 10^{-5}$).

Quantity checked	**PyTorch reference**	**Result**
Adam iterate x_t	`optim.Adam`	Pass
AMSGrad iterate x_t	`Adam(amsgrad=True)`	Pass
First moment m_t	`exp_avg`	Pass
Second moment v_t	`exp_avg_sq`	Pass
Running max $\hat{v}_t$	`max_exp_avg_sq`	Pass

References

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